

Scholara

Open Knowledge, Everywhere

User Manual & Functional Verification Guide

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About This Guide

This manual walks you through every feature of Scholara step by step. Each section includes an easy example you can follow right now and a clear 'What to verify' checklist.

You do not need any technical background. All you need is Scholara running on your computer and a web browser.

What is Scholara?

Scholara is a free, open-source academic research platform designed for researchers and students in Africa and the Global South. It lets you:

- Search millions of open-access papers, books, and journals with a single query.
- Download academic PDFs and documents directly to your computer, for free.
- Convert files between formats (PDF → Word, Word → PDF, video → audio).
- Organise your research with history tracking, tags, and named collections.
- Export citations in BibTeX, RIS, and APA formats.
- Use the interactive AI demo to explain, summarise, and chat with any document.
- Use the interface in English or French.
- Install as an app on your phone or desktop (PWA).

Component	Technology
Backend	Python 3.14 + FastAPI
Frontend	Vanilla HTML/CSS/JS — ES modules, no framework
Database	SQLite (scholara.db)
AI Router	Ollama (local) → DeepSeek (cloud) → Keyword fallback
Download	Direct HTTP — open-access domains only
Conversion	Pure-Python: pdf2docx, pypdf, mammoth, ffmpeg
Auth (multi-user)	python-jose[cryptography] · passlib[bcrypt]

Section 1 — Getting Started

1.1 Starting the Application

Open a terminal, navigate to the Scholara folder, and run:

```
# First time only
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt

# Every time
python app.py
```

Scholara opens automatically at: <http://127.0.0.1:7860>

1.2 The Status Banner

```
=== Scholara Status ===
[OK] Ollama (mistral)
[--] DeepSeek API
[OK] ffmpeg (conversion)
[OK] Direct HTTP: always available
=====
```

TIP: If Ollama shows [--], the AI router falls back to keyword matching. Search still works.

1.3 The Interface at a Glance

Area	What it does
Header bar	Logo, language toggle (EN / FR), status badges, Sign Out button
Left sidebar tabs	Search · URL · Convert · History · Collections · Admin
Results grid	Cards showing search results with Download buttons
Right viewer	Inline preview of PDFs, text, audio, and video files
Demo sidebar	Slides in from the right — AI explanation, summary, chat, slides, flowchart
Settings modal	API keys, download folder, dark mode, language preference

Section 2 — Natural Language Search

Type a question or topic in plain English or French. The AI router selects the most relevant sources automatically. Results are deduplicated and reranked by relevance.

2.1 English Search — Example

- Click the Search tab in the left sidebar.
- Type: machine learning applications in African agriculture
- Press Enter or click Search.

What to verify:

- Results appear within 10 seconds.
- At least one result from arXiv (📄) or OpenAlex (🔍).
- Each card shows: Title, Authors, Source badge, Download button.

2.2 French Search — Example

histoire de l'éducation en Afrique subsaharienne

What to verify:

- HAL (🏛️) results appear — France's national open archive.
- Routing label shows 'hal' and/or 'persee' as selected sources.

TIP: Scholara detects French queries and automatically prioritises HAL, Persée, OpenEdition, and Érudit.

2.3 Search Intelligence

Feature	How it works
Deduplication	Removes near-duplicate results — exact DOI match or >90% title similarity
Reranking	+3 language match · +2 DOI present · +2 abstract · +1 recent · +1 open-access

2.4 Core and Global North Sources

Source	Language	Always active?
arXiv	EN	Yes
DOAJ	EN/FR	Yes
OpenAlex	EN	Yes
Internet Archive	EN	Yes
Project Gutenberg	EN	Yes
HAL	FR/EN	Yes
Persée	FR	Yes
OpenEdition	FR/EN	Yes
Érudit	FR/EN	Yes
Semantic Scholar	EN	Global North only
PubMed	EN	Global North only
CrossRef	EN	Global North only
CORE	EN	Global North only
BASE	Multilingual	Global North only

Section 3 — Downloading Files

3.1 Download from Search Results

- Run any search — click 'Download PDF' on any result card.

What to verify:

- Progress bar appears → file appears in History and file list when done.
- Clicking the file opens it in the viewer.

3.2 Download by Pasting a URL

- Click the URL tab — paste an open-access URL:

`https://arxiv.org/pdf/1706.03762.pdf`

- Click Download.

What to verify:

- Status: queued → running → done.
- File appears in file list and History.

3.3 Blocked URL Test

- Paste: `https://www.google.com/file.pdf` — Click Download.

What to verify:

- Error message appears — no file created. Only whitelisted open-access domains are allowed.

Section 4 — Interactive AI Demo

The Interactive Demo lets you analyse any downloaded file or selected text with five AI-powered actions. Right-click a file entry, or select text anywhere on the page to trigger it.

4.1 How to Trigger the Demo

Trigger	How
Right-click a file	Right-click any entry in the file list or History tab
Text selection	Select more than 20 characters anywhere → context menu appears

4.2 Five AI Actions

Action	What it produces
 Text Explanation	Plain-language explanation for a non-specialist reader
 Summary	Structured: main topic · key findings · methodology · limitations
 Interactive Demo	Multi-turn AI chat grounded in the document text
 Presentation	Slide-by-slide outline rendered as cards in the sidebar
 Flowchart	Mermaid.js diagram of the main process or argument

4.3 Text Explanation — Example

- Download arXiv paper 1706.03762.pdf (Section 3.2).
- Right-click the file entry → select ' Text Explanation'.

What to verify:

- Demo sidebar slides in from the right.
- Loading spinner appears briefly, then the explanation renders.
- Response language matches your current interface language (EN or FR).

4.4 Interactive Chat — Example

- Right-click any PDF → ' Interactive Demo'.
- Sidebar opens with an introductory response and a chat input.
- Type a follow-up question and press Enter:

```
What is the main contribution of this paper?
```

What to verify:

- Response appears, grounded in the document.
- Up to 6 turns of history are maintained.
- Shift+Enter for a new line; Enter sends the message.

4.5 Presentation — Example

- Select a large block of abstract text → ' Presentation'.

What to verify:

- 4–8 slide cards appear, each with title and bullets.
- If AI returns invalid JSON, raw text is shown instead (graceful fallback).

4.6 Flowchart — Example

- Right-click a methods-section PDF → ' Flowchart'.

What to verify:

- Mermaid.js diagram renders in the sidebar.
- If Mermaid fails to parse, raw text is shown instead.

NOTE: Mermaid.js loads from CDN on first use — internet required for the first flowchart.

4.7 Offline AI Fallback

If no AI backend is available, all demo actions return a graceful message:

```
AI backend unavailable. Please start Ollama or configure a DeepSeek API key.
```

4.8 Supported File Types

Extension	Extraction method
.pdf	pypdf — text layer extracted page by page
.docx	mammoth — raw text extraction
.txt / .md	Direct file read
.html	html2text — Markdown-like plain text

Section 5 — File Conversion

From	To	Requires
PDF	Word (.docx), plain text, HTML	pdf2docx, pypdf (included)
Word (.docx)	PDF, HTML, Markdown, plain text	mammoth (included)
Markdown / HTML / text	PDF, Word, HTML	included
Video (mp4, webm...)	MP3, WAV, AAC, OGG, FLAC	system ffmpeg

5.1 Convert PDF to Word

- Click Convert tab → select PDF → choose 'docx' → click Convert.

What to verify:

- New file '1706.03762_converted.docx' appears. Clicking it opens the Word document.

5.2 Convert PDF to Text

- Same PDF → format 'txt' → Convert.

What to verify:

- File '1706.03762_converted.txt' appears and shows plain text in the viewer.

Section 6 — History & Collections

6.1 Viewing History

What to verify:

- Each entry shows: Filename, Source, Date, File size.
- Clicking an entry opens the file in the right viewer.
- Missing files flagged in red.

6.2 Create a Collection

- Collections tab — 'New Collection' — enter name.
- Optionally tick 'Share with institution' (multi-user mode).
- Click Create.

What to verify:

- Collection appears in the list.
- Shared collections are accessible at GET /api/collections/shared.

6.3 Add a Paper to a Collection

- In History, click the  button on any paper — select a collection.

What to verify:

- Opening the collection shows the paper inside it.

Section 7 — Citation Export

```
GET /api/cite/1?format=bibtex
```

```
GET /api/cite/1?format=ris
```

```
GET /api/cite/1?format=apa
```

TIP: Replace '1' with the actual history entry ID shown in the History tab.

7.1 BibTeX

What to verify:

- Browser downloads a .bib file starting with @article{scholara:...
- Contains title, author, year, journal, url fields.

7.2 RIS

What to verify:

- Content starts with 'TY - JOUR' and ends with 'ER -'.
- File imports directly into Zotero or Mendeley.

7.3 APA

What to verify:

- Plain text in APA style: Author (Year). Title. Source. URL

Section 8 — Bilingual Interface (English / French)

French is the default language. Click the EN/FR button in the header, or open Settings.

What to verify:

- All tab labels, buttons, placeholders, and messages switch instantly.
- Language choice persists across page reloads.

Element	English	French
Page tagline	Open Knowledge, Everywhere	Le Savoir Ouvert, Partout
Search tab	 Search	 Recherche
History tab	 History	 Historique
Demo: Explanation	Text Explanation	Explication textuelle
Demo: Chat	Interactive Demo	Démo interactive
Demo: Flowchart	Flowchart	Organigramme

Section 9 — Settings

Setting	What it does
Download directory	Where files are saved on disk
Max concurrent	Parallel downloads (1-5)
Dark mode	Toggle dark theme
Language	Interface language preference
Auto-open viewer	Open files in viewer after download

9.1 Check System Status

`http://127.0.0.1:7860/api/status`

What to verify:

- JSON shows `ollama_available`, `market_segment`, `active_sources`, `app_mode`.
- `institution_branding` shows `logo_url` / `primary_color` when set (multi-user).

Section 10 — Multi-User Mode

Multi-user mode adds JWT-based login, role management, shared institutional collections, usage analytics, and an audit log.

10.1 Enabling Multi-User Mode

```
APP_MODE=multi_user SECRET_KEY=your-secret python app.py
```

NOTE: Change SECRET_KEY to a long random string in production.

10.2 User Roles

Role	Permissions
admin	Full access: users, audit log, roles, impact report, shared collections
researcher	Search, download, convert, own history and collections
student	Same as researcher

10.3 Registering and Logging In

```
# Register
curl -X POST http://127.0.0.1:7860/api/auth/register \
  -H 'Content-Type: application/json' \
  -d '{"email":"you@example.com","password":"s3cr3t"}'

# Login
curl -X POST http://127.0.0.1:7860/api/auth/login \
  -H 'Content-Type: application/json' \
  -d '{"email":"you@example.com","password":"s3cr3t"}'
```

10.4 Admin Impact Report

GET /api/admin/impact returns full analytics:

Field	Description
total_downloads	Total number of files downloaded
total_users	Total registered users
sources_used	Download count per source, sorted descending
downloads_by_day	Download counts for the last 30 days
top_queries	Top 10 anonymised search query stems with counts
top_sources	Top 5 most-used sources
active_users_week	Distinct users active in the last 7 days

10.5 Institutional Branding

Set a logo_url and primary_color for your institution. The frontend applies the primary_color as a CSS variable, changing the accent colour throughout the interface.

```
POST /api/admin/institutions
{"name": "Université de Dakar",
 "country": "SN",
 "logo_url": "https://example.edu/logo.png",
 "primary_color": "#0057a8"}
```

10.6 Shared Collections

Share any collection with all users in your institution. The 'Share with institution' checkbox in the New Collection modal does this automatically.

```
POST /api/collections/{id}/share
GET  /api/collections/shared
```

Section 11 — Admin Analytics Tab

Admin users see a  Admin tab in the sidebar with a live analytics panel and a bar chart of downloads by day.

What to verify:

- Total downloads and total users are shown.
- 'Active this week' shows distinct users active in the last 7 days.
- Top 5 search queries with counts (anonymised query stems).
- Top sources appear with download counts.
- Bar chart shows downloads per day for the last 30 days.

Section 12 — Install as an App (PWA)

12.1 Install on Desktop

- Open Scholarla in Chrome or Edge.
- Click the  icon in the address bar → 'Install Scholarla'.

12.2 Install on Android

- Phone and computer on the same Wi-Fi.
- Open Chrome on Android → `http://192.168.1.x:7860`
- Three-dot menu → 'Add to Home screen'.

NOTE: Edit the last line of `app.py` to use `host='0.0.0.0'` to serve on your local network.

Section 13 — Running with Docker

```
# Start with Ollama sidecar
docker-compose up --build
```

```
# Multi-user with Docker
APP_MODE=multi_user SECRET_KEY=your-secret docker-compose up
```

What to verify:

- Scholarla loads at `http://localhost:7860`
- `/api/status` shows 'ollama_available': true.

Section 14 — Quick Verification Checklist

#	Feature	Test	Pass criterion
1	App loads	Open http://127.0.0.1:7860	Scholara interface appears
2	Status API	GET /api/status	JSON with tool flags
3	English search	Search: 'climate change models'	arXiv / OpenAlex results
4	French search	Search: 'changement climatique Afrique'	HAL results appear
5	Direct download	URL: arxiv.org/pdf/1706.03762.pdf	File saved; history entry created
6	Blocked URL	URL: google.com/file.pdf	Error; no file created
7	PDF → Word	Convert PDF to .docx	New _converted.docx in list
8	History view	Click History tab	Downloads listed with date/size
9	Create collection	Create 'My Test Collection'	Appears in Collections tab
10	Add to collection	Add a paper to the collection	Paper shows inside collection
11	BibTeX export	GET /api/cite/1?format=bibtex	@article{...} output returned
12	Language toggle	Click EN/FR button	All labels switch instantly
13	Demo — explain	Right-click PDF →  Text Explanation	Sidebar opens with explanation
14	Demo — summary	Right-click PDF →  Summary	Structured summary appears
15	Demo — chat	Right-click PDF →  Interactive Demo	Chat with follow-ups works
16	Demo — slides	Select text →  Presentation	Slide cards rendered
17	Demo — flowchart	Select text →  Flowchart	Mermaid diagram rendered
18	Demo fallback	Disable Ollama; try any action	Graceful canned message shown
19	Demo on selection	Select >20 chars → menu appears	Context menu shown
20	Search dedup	Search with duplicate sources	X-Search-Deduped header present
21	Admin analytics tab	Click  Admin tab	Bar chart + stats render
22	Shared collection	Create collection with share ticked	Visible at /api/collections/shared
23	Institution branding	POST institution with primary_color	Accent colour changes on reload
24	Multi-user login	POST /api/auth/login (multi_user mode)	access_token + refresh_token returned
25	Admin impact	GET /api/admin/impact (admin token)	downloads_by_day, top_queries, active_users_week present
26	PWA install	Open in Chrome; look for 	Install prompt appears
27	Queue persistence	Start download; restart app; check queue	Job re-appears as queued

Section 15 — Troubleshooting

The app does not open in the browser:

Make sure python app.py is running. Open `http://127.0.0.1:7860` manually.

Search returns no results:

Check internet connection. All search sources require internet access.

AI search says 'fallback' in routing info:

Ollama is not running. Run 'ollama serve' and 'ollama pull mistral'.

Demo sidebar shows 'AI backend unavailable':

Same as above — start Ollama or set `DEEPSEEK_API_KEY`. The feature degrades gracefully.

Demo sidebar does not open for my file:

Only .pdf, .docx, .txt, .md, and .html files are supported. Check the file extension.

Flowchart does not render:

Mermaid.js loads from CDN on first use — internet required. If AI returned invalid syntax, raw text is shown.

PDF conversion fails:

Very scanned PDFs (image-only) cannot be converted to text or Word.

Video conversion fails:

ffmpeg must be installed: `sudo apt install ffmpeg` (Linux) or `brew install ffmpeg` (macOS).

Language does not switch:

Clear localStorage: DevTools — Application — Local Storage — delete 'lang' key.

Login returns 401 with correct credentials:

Set `APP_MODE=multi_user` and restart. In `single_user` mode no login endpoint is active.

Access token expired — 401 on API calls:

Use `POST /api/auth/refresh` with your `refresh_token` (valid 7 days) to get a new `access_token`.

Admin routes return 403:

Your account has role 'researcher'. Use `PATCH /api/admin/users/{id}/role` (admin token) to promote it.

Shared collections not visible:

Users must have the same `institution_id`. Check both accounts with `GET /api/admin/users`.

Admin analytics tab not showing:

The tab is hidden until an admin role is detected. In single-user mode it always shows after page load.

CORE search returns no results:

Set `CORE_API_KEY=your-key`. Without the key, CORE silently returns an empty list.

Scholara

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arXiv · DOAJ · OpenAlex · Project Gutenberg · Internet Archive · HAL · Persée · OpenEdition · Érudit

Global North: Semantic Scholar · PubMed · CrossRef · CORE · BASE